

Campus Event and Club Management Platform

Functional Requirements

These describe what the system must do.

A. Student

The system should allow a student to:

1. Register/login to the system.
2. View upcoming campus events.
3. View event details.
4. Filter events by:
 - Category
 - Date
5. Search for events.
6. RSVP for an event.
7. Cancel an RSVP.
8. See whether an event has reached its capacity.
9. Receive/generate a QR code ticket after RSVP.
10. View their registered events.
11. Use the QR ticket for event check-in.

B. Club President / Club Organizer

The system should allow a club president to:

1. Login as a club president.
2. Create an event.
3. Edit an event.
4. Delete/cancel an event.
5. Add event information:
 - Event name
 - Description
 - Date
 - Time
 - Location

- Category
 - Capacity
6. View the number of RSVPs.
 7. Track event capacity.
 8. View registered participants.
 9. Check attendance.
 10. View event engagement information.

C. Admin

The system should allow an administrator to:

1. Login to the admin panel.
2. View submitted events.
3. Approve events.
4. Reject events.
5. Manage events/users/clubs if required.
6. View engagement statistics.
7. View event analytics.
8. Monitor attendance.

D. QR Ticket & Check-in

The system should:

1. Generate a unique QR code for an RSVP.
2. Associate the QR code with a student and event.
3. Allow staff/club members to scan the QR code.
4. Validate the QR code.
5. Prevent duplicate check-ins.
6. Record check-in time.
7. Show whether the ticket is valid/invalid.

E. Analytics

The system should provide information such as:

- Total events
- Total RSVPs
- Total attendees

- Attendance rate
- Most popular events
- Event engagement
- Events by category
- Registration trends

Non-Functional Requirements

These describe how well the system should work.

Performance

- Event pages should load quickly.
- QR validation should respond quickly.
- The system should handle multiple users.

Security

- Users should authenticate before accessing protected features.
- Different users should have different permissions.
- Students should not access admin functions.
- QR tickets should be unique.
- Passwords should not be stored as plain text.

Usability

- Interface should be simple.
- Students should easily find events.
- RSVP should require only a few steps.
- QR check-in should be easy to use.

Reliability

- RSVP data should not be lost.
- Capacity should be updated correctly.
- Duplicate RSVPs should not occur.
- Duplicate check-ins should be prevented.

Maintainability

- Code should be organized.
- Git/GitHub should be used.
- The project should have documentation.

Compatibility

The system should work on common:

- Desktop browsers
- Mobile browsers